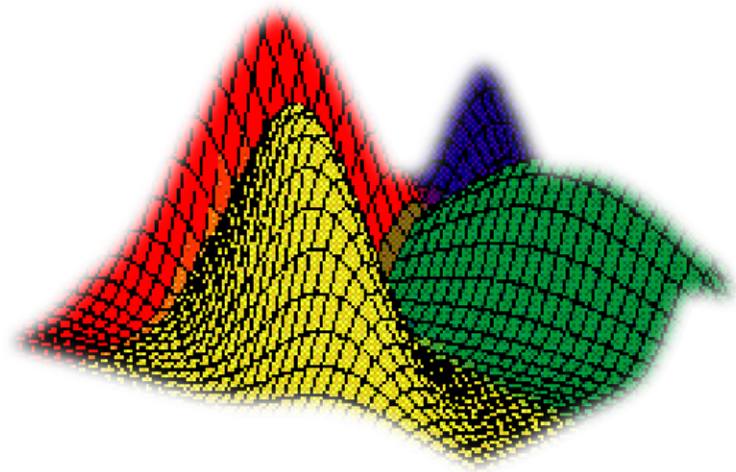




Bayesian Network

Graphical Models

- Key Idea:
 - Conditional independence assumptions useful
 - but Naïve Bayes is extreme!
 - Graphical models express sets of conditional independence assumptions via graph structure
 - Graph structure plus associated parameters define joint probability distribution over set of variables

Graphical Models – Why Care?

- Among most important ML developments of the decade
- Graphical models allow combining:
 - Prior knowledge in form of dependencies/independencies
 - Prior knowledge in form of priors over parameters
 - Observed training data
- Principled and ~general methods for
 - Probabilistic inference
 - Learning
- Useful in practice
 - Diagnosis, help systems, text analysis, time series models, ...

Conditional Independence

Definition: X is conditionally independent of Y given Z, if the probability distribution governing X is independent of the value of Y, given the value of Z

$$(\forall i, j, k) P(X = x_i | Y = y_j, Z = z_k) = P(X = x_i | Z = z_k)$$

Which we often write $P(X|Y, Z) = P(X|Z)$

E.g., $P(\text{Thunder} | \text{Rain}, \text{Lightning}) = P(\text{Thunder} | \text{Lightning})$

Marginal Independence

Definition: X is marginally independent of Y if

$$(\forall i, j) P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j)$$

Equivalently, if

$$(\forall i, j) P(X = x_i | Y = y_j) = P(X = x_i)$$

Equivalently, if

$$(\forall i, j) P(Y = y_i | X = x_j) = P(Y = y_i)$$

Bayes Nets define Joint Probability Distribution in terms of this graph, plus parameters

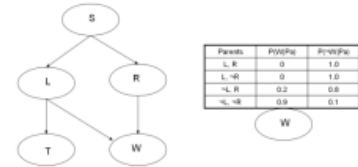


$$P(X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8) \\ = P(X_1) P(X_2) P(X_3 | X_1) P(X_4 | X_2) P(X_5 | X_2) \\ P(X_6 | X_3, X_4) P(X_7 | X_6) P(X_8 | X_5, X_6)$$

Benefits of Bayes Nets:

- Represent the full joint distribution in fewer parameters, using prior knowledge about dependencies
- Algorithms for inference and learning

Bayesian Networks Definition



A Bayes network represents the joint probability distribution over a collection of random variables

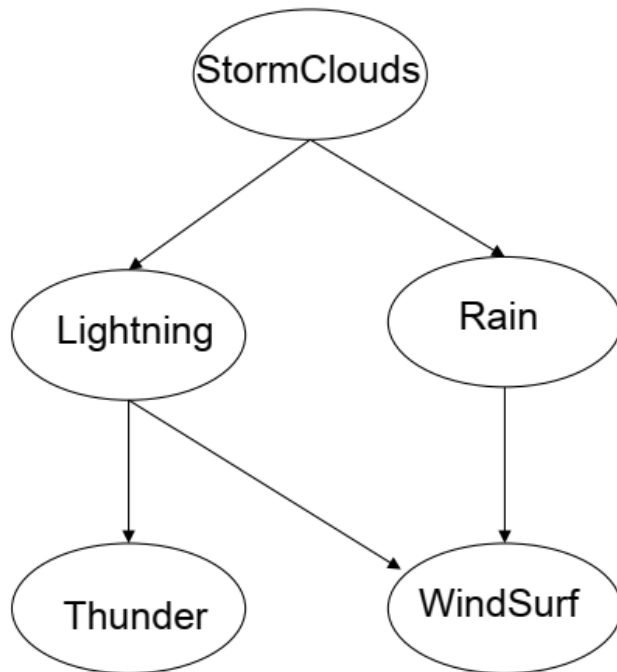
A Bayes network is a directed acyclic graph and a set of conditional probability distributions (CPD's)

- Each node denotes a random variable
- Edges denote dependencies
- For each node X_i its CPD defines $P(X_i | Pa(X_i))$
- The joint distribution over all variables is defined to be

$$P(X_1 \dots X_n) = \prod_i P(X_i | Pa(X_i))$$

$Pa(X)$ = immediate parents of X in the graph

Bayesian Network



Nodes = random variables

A conditional probability distribution (CPD) is associated with each node N , defining $P(N \mid \text{Parents}(N))$

Parents	$P(W Pa)$	$P(\neg W Pa)$
L, R	0	1.0
L, $\neg R$	0	1.0
$\neg L$, R	0.2	0.8
$\neg L$, $\neg R$	0.9	0.1



The joint distribution over all variables:

$$P(X_1 \dots X_n) = \prod_i P(X_i | Pa(X_i))$$

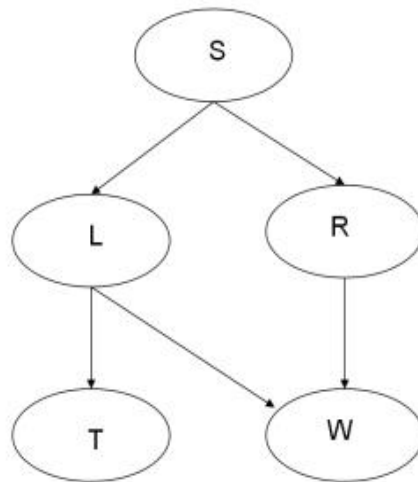
Some helpful terminology

Parents = $\text{Pa}(X)$ = immediate parents

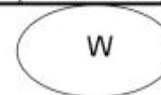
Antecedents = parents, parents of parents, ...

Children = immediate children

Descendents = children, children of children, ...



Parents	$P(W \text{Pa})$	$P(\neg W \text{Pa})$
L, R	0	1.0
L, $\neg R$	0	1.0
$\neg L$, R	0.2	0.8
$\neg L$, $\neg R$	0.9	0.1

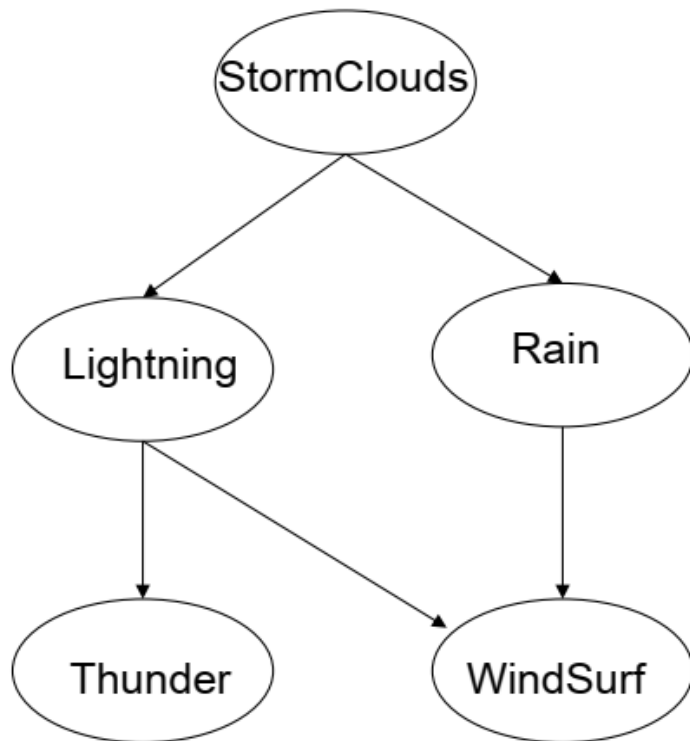


Bayesian Network

What can we say about conditional independencies in a Bayes Net?

One thing is this:

Each node is conditionally independent of its non-descendents, given only its immediate parents.

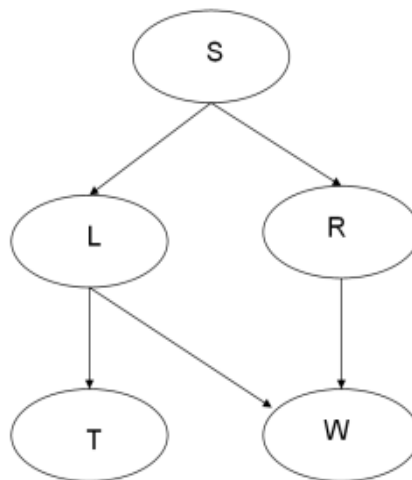


Parents	$P(W Pa)$	$P(\neg W Pa)$
L, R	0	1.0
L, $\neg R$	0	1.0
$\neg L$, R	0.2	0.8
$\neg L$, $\neg R$	0.9	0.1

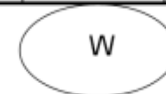


Bayesian Networks

- CPD for each node X_i describes $P(X_i | Pa(X_i))$



Parents	$P(W Pa)$	$P(\neg W Pa)$
L, R	0	1.0
L, $\neg R$	0	1.0
$\neg L$, R	0.2	0.8
$\neg L$, $\neg R$	0.9	0.1



Chain rule of probability says that in general:

$$P(S, L, R, T, W) = P(S)P(L|S)P(R|S, L)P(T|S, L, R)P(W|S, L, R, T)$$

8 params

But in a Bayes net: $P(X_1 \dots X_n) = \prod_i P(X_i | Pa(X_i))$

$$P(S, L, R, T, W) = P(S) P(L|S) P(R|S) P(T|L) P(W|L, R)$$

2

Bayesian networks

a BN consists of a Directed Acyclic Graph (DAG) and a set of conditional probability distributions

in the DAG

- each node denotes random a variable
- each edge from X to Y represents that X *directly influences* Y
- formally: each variable X is independent of its non-descendants given its parents

each node X has a *conditional probability distribution* (CPD) representing $P(X \mid Parents(X))$

Bayesian networks

Using the chain rule, a joint probability distribution can be expressed as

$$P(X_1, \dots, X_n) = P(X_1) \prod_{i=2}^n P(X_i \mid X_1, \dots, X_{i-1}))$$

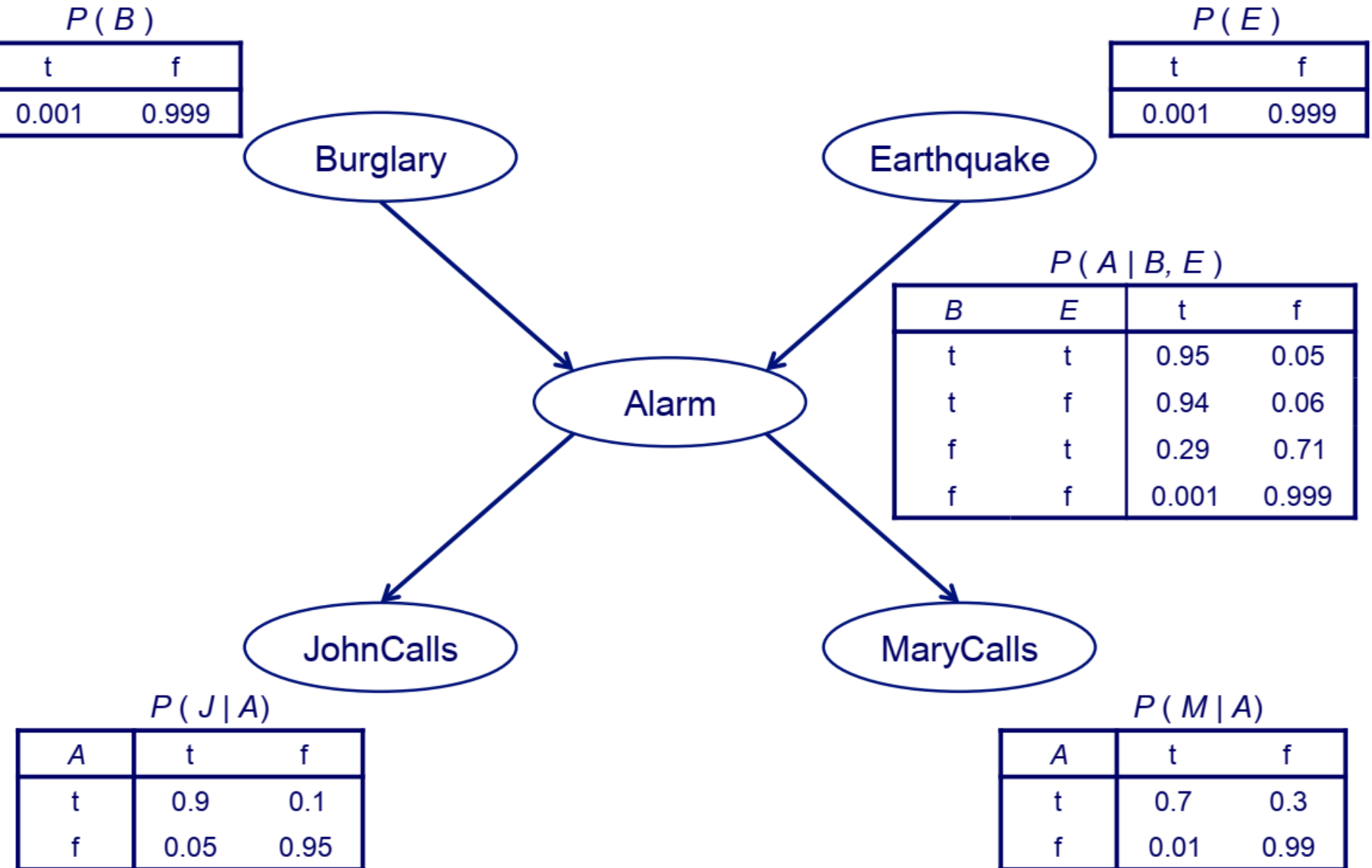
BN provides a compact representation of a joint probability distribution

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$$

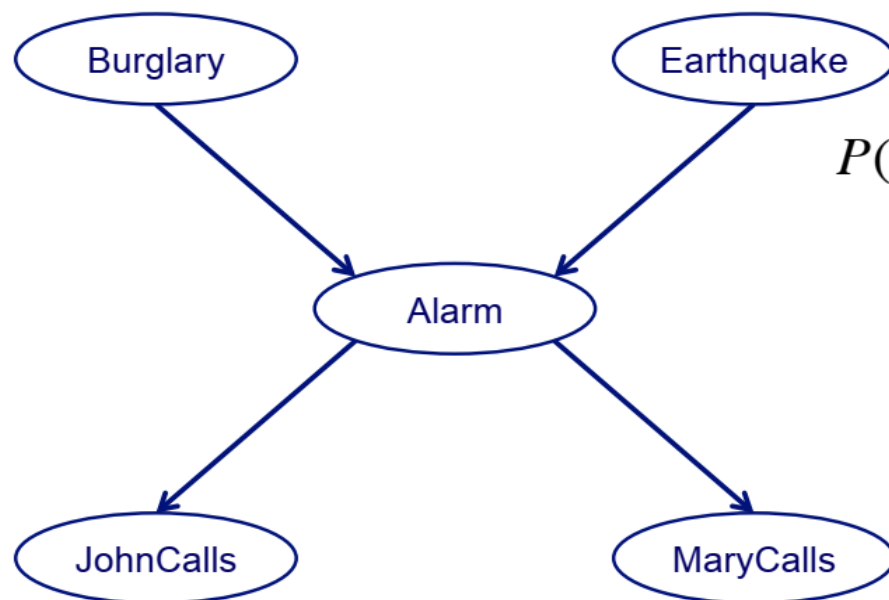
Bayesian network example

- Consider the following 5 binary random variables:
 - B = a burglary occurs at your house
 - E = an earthquake occurs at your house
 - A = the alarm goes off
 - J = John calls to report the alarm
 - M = Mary calls to report the alarm
- Suppose we want to answer queries like what is

Bayesian network example



Bayesian networks

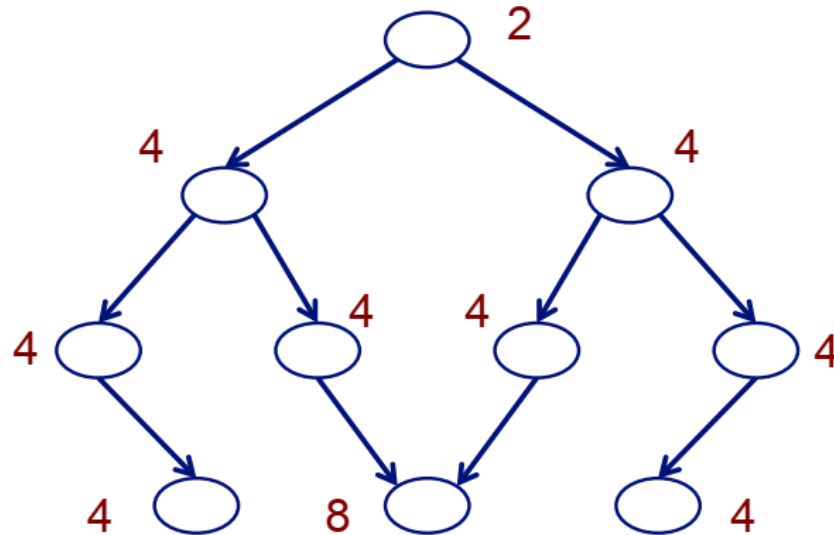


$$P(B, E, A, J, M) = P(B) \times P(E) \times P(A | B, E) \times P(J | A) \times P(M | A)$$

- a standard representation of the joint distribution for the Alarm example has $2^5 = 32$ parameters
- the BN representation of this distribution has 20 parameters

Bayesian networks

- consider a case with 10 binary random variables
- How many parameters does a BN with the following graph structure have?



= 42

- How many parameters does the standard table representation of the joint distribution have? = 1024

Advantages of the Bayesian network representation

Captures independence and conditional independence where they exist

Encodes the relevant portion of the full joint among variables where dependencies exist

Uses a graphical representation which lends insight into the complexity of inference

Bayesian Belief Networks

- Let we l random variables
- The joint probability is given by,

$$p(x_1, x_2, \dots, x_\ell) = p(x_\ell \mid x_{\ell-1}, \dots, x_1) \cdot p(x_{\ell-1} \mid x_{\ell-2}, \dots, x_1) \cdot \dots \\ \dots \cdot p(x_2 \mid x_1) \cdot p(x_1)$$

Bayesian Belief Networks

The formula

$$p(x_1, x_2, \dots, x_\ell) = p(x_\ell \mid x_{\ell-1}, \dots, x_1) \cdot p(x_{\ell-1} \mid x_{\ell-2}, \dots, x_1) \cdot \dots \\ \dots \cdot p(x_2 \mid x_1) \cdot p(x_1)$$

can be written as

$$p(x_1, x_2, \dots, x_\ell) = p(x_1) \cdot \prod_{i=2}^{\ell} p(x_i \mid A_i)$$

where

$$A_i \subseteq \{x_{i-1}, x_{i-2}, \dots, x_1\}$$

- For example, if $\ell=6$, then we could assume:

$$p(x_6 \mid x_5, \dots, x_1) = p(x_6 \mid x_5, x_4)$$

Then:

$$A_6 = \{x_5, x_4\} \subseteq \{x_5, \dots, x_1\}$$

– Similarly, if we assume

$$p(x_5|x_4, \dots, x_1) = p(x_5|x_4)$$

$$p(x_4|x_3, x_2, x_1) = p(x_4|x_2, x_1)$$

$$p(x_3|x_2, x_1) = p(x_3|x_2)$$

$$p(x_2|x_1) = p(x_2)$$

Then:

$$A_5 = \{x_4\}, A_4 = \{x_2, x_1\}, A_3 = \{x_2\}, A_2 = \emptyset$$

- Similarly, if we assume

$$p(x_5|x_4, \dots, x_1) = p(x_5|x_4)$$

$$p(x_4|x_3, x_2, x_1) = p(x_4|x_2, x_1)$$

$$p(x_3|x_2, x_1) = p(x_3|x_2)$$

$$p(x_2|x_1) = p(x_2)$$

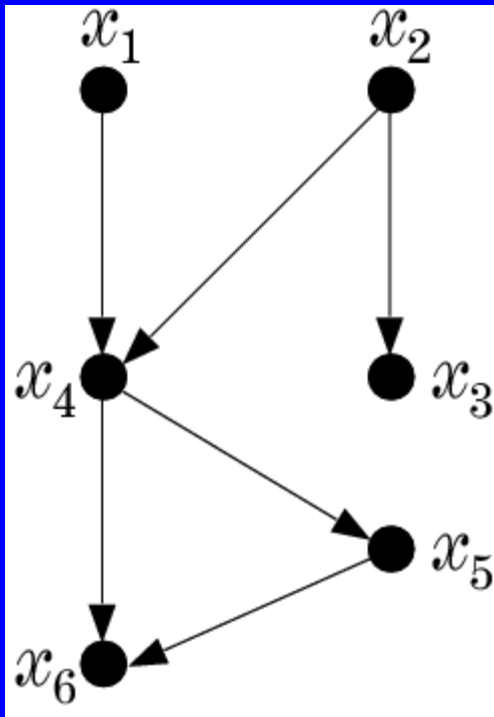
Then:

$$A_5 = \{x_4\}, A_4 = \{x_2, x_1\}, A_3 = \{x_2\}, A_2 = \emptyset$$

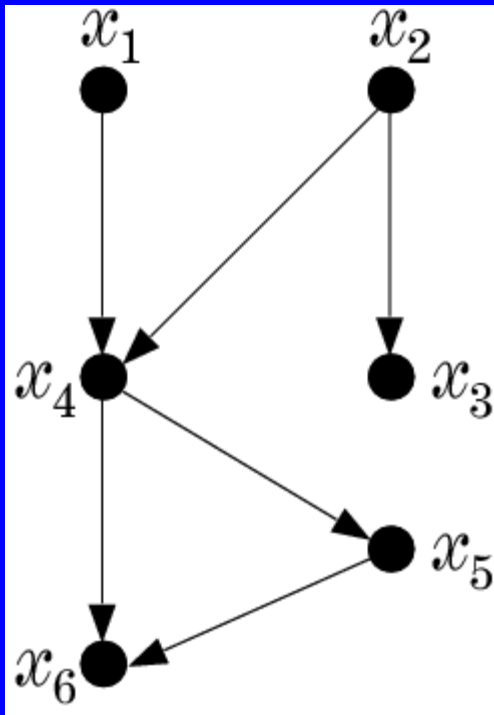
- The above is a generalization of the Naïve – Bayes. For the Naïve – Bayes the assumption is:

$$A_i = \emptyset, \text{ for } i=1, 2, \dots, \ell$$

- A graphical way to portray conditional dependencies



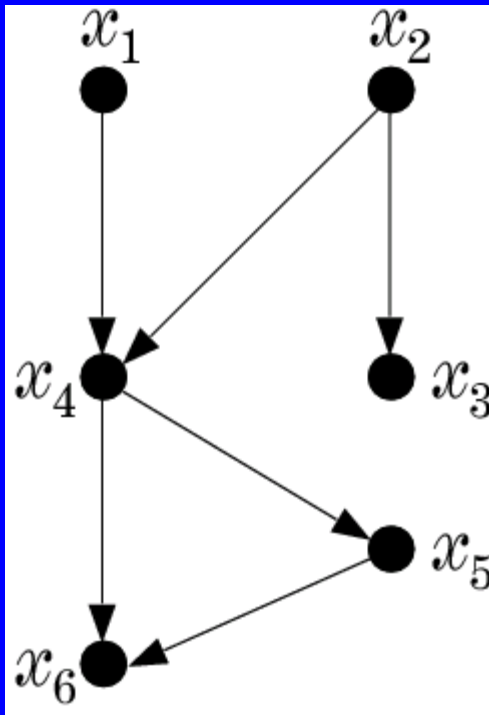
- A graphical way to portray conditional dependencies



➤ According to this figure we have that:

- x_6 is conditionally dependent on x_4, x_5 .
- x_5 on x_4
- x_4 on x_1, x_2
- x_3 on x_2
- x_1, x_2 are conditionally independent on other variables.

- A graphical way to portray conditional dependencies



➤ According to this figure we have that:

- x_6 is conditionally dependent on x_4, x_5 .
- x_5 on x_4
- x_4 on x_1, x_2
- x_3 on x_2
- x_1, x_2 are conditionally independent on other variables.

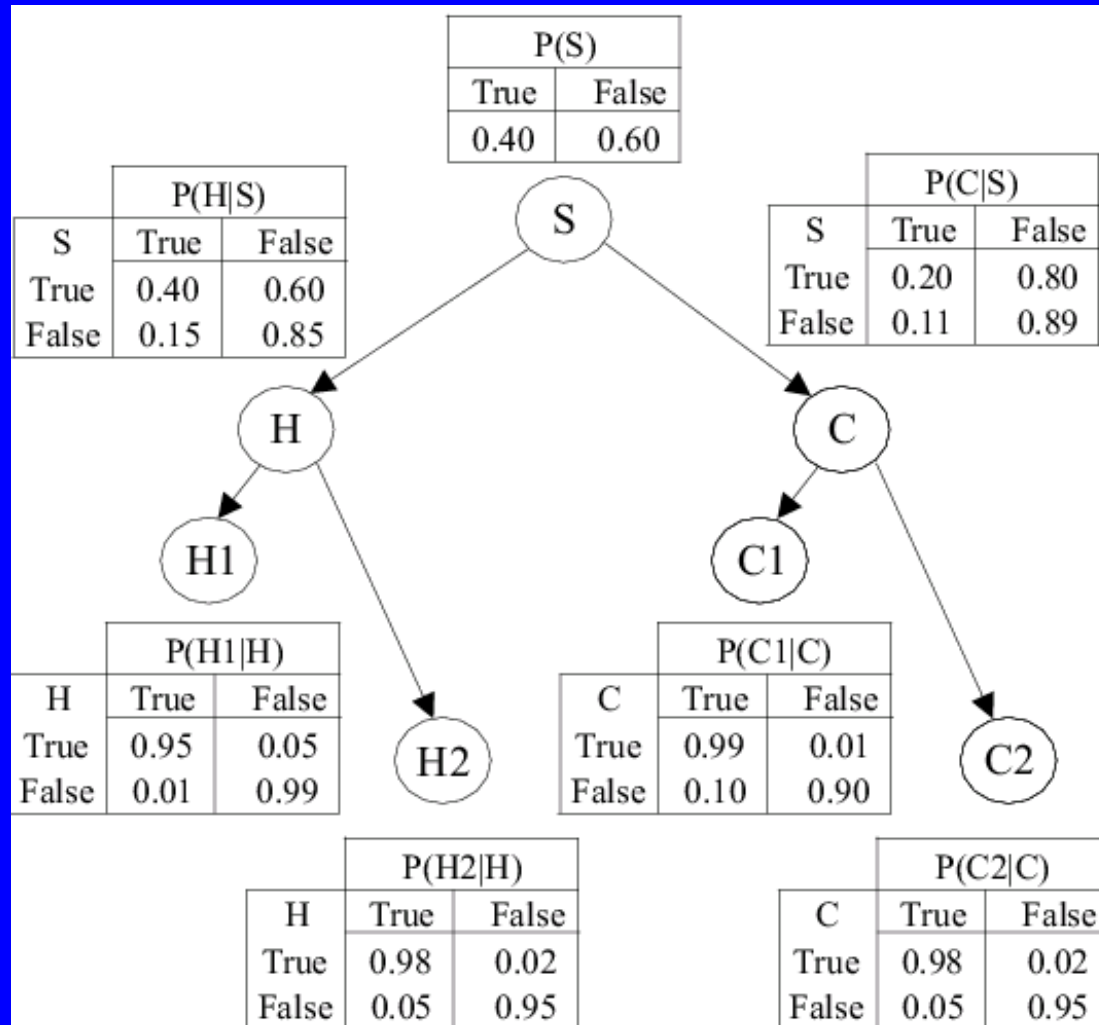
➤ For this case:

$$p(x_1, x_2, \dots, x_6) = p(x_6 \mid x_5, x_4) \cdot p(x_5 \mid x_4) \cdot p(x_3 \mid x_2) \cdot p(x_2) \cdot p(x_1)$$

- Bayesian Networks
 - a directed acyclic graph (DAG)
 - the nodes correspond to random variables
 - arc represents parent-child (*dependence*) relationship

- A Bayesian Network is specified by:
 - The prior probabilities of its root nodes.
 - The conditional probabilities of the non-root nodes, given their parents, for ALL possible combinations.

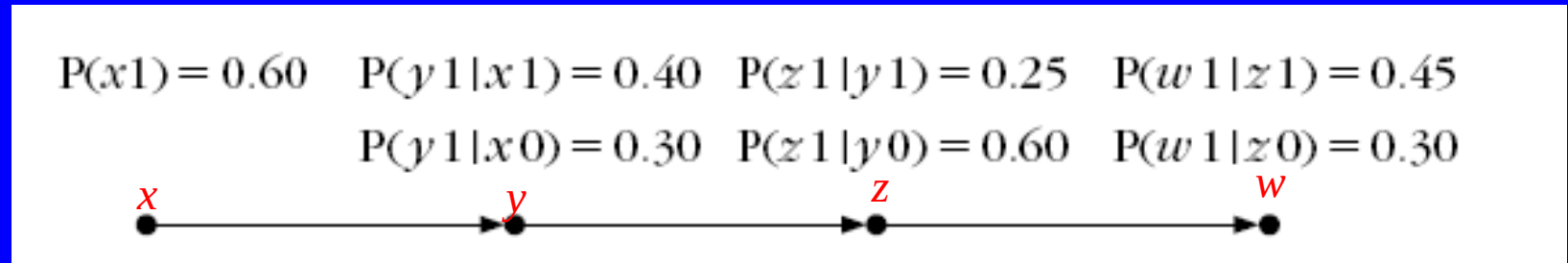
– A Bayesian Network from a medical application



➤ BBN models conditional dependencies concerning smokers (S), tendencies to develop cancer (C) and heart disease (H), together with variables corresponding to heart (H1, H2) and cancer (C1, C2) medical tests.

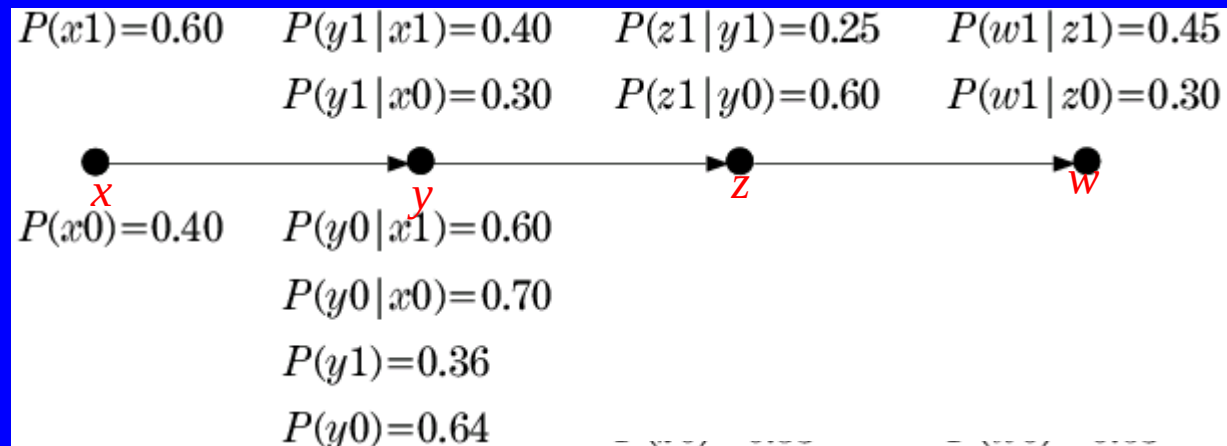
- any joint probability can be obtained by multiplying the prior (root nodes) and the conditional (non-root nodes) probabilities.
- **Training:** given a topology, probabilities are estimated from training data. There are also methods that learn the topology.
- **Probability Inference:** Given an pattern (evidence), the goal is to compute the conditional probabilities for some of the other variables (class)

- Example: Consider the Bayesian network of the figure:



- Random variables: x, y, w, z
- x_0 means $x = 0$
- x_1 means $x = 1$

- We can calculate the other probabilities



Example: $p(y_1)$:

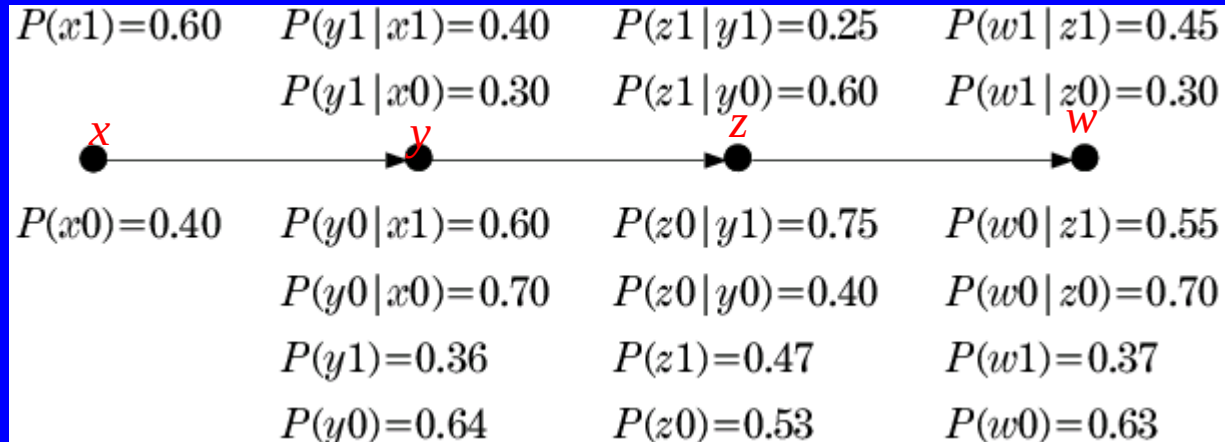
$$P(y1) = \sum_x P(y1, x) = P(y1, x1) + P(y1, x0)$$

$$P(y1) = P(y1|x1)P(x1) + P(y1|x0)P(x0) = (0.4)(0.6) + (0.3)(0.4) = 0.36$$

- We can calculate the other probabilities

$P(x1)=0.60$	$P(y1 x1)=0.40$	$P(z1 y1)=0.25$	$P(w1 z1)=0.45$
	$P(y1 x0)=0.30$	$P(z1 y0)=0.60$	$P(w1 z0)=0.30$
			
$P(x0)=0.40$	$P(y0 x1)=0.60$	$P(z0 y1)=0.75$	$P(w0 z1)=0.55$
	$P(y0 x0)=0.70$	$P(z0 y0)=0.40$	$P(w0 z0)=0.70$
	$P(y1)=0.36$	$P(z1)=0.47$	$P(w1)=0.37$
	$P(y0)=0.64$	$P(z0)=0.53$	$P(w0)=0.63$

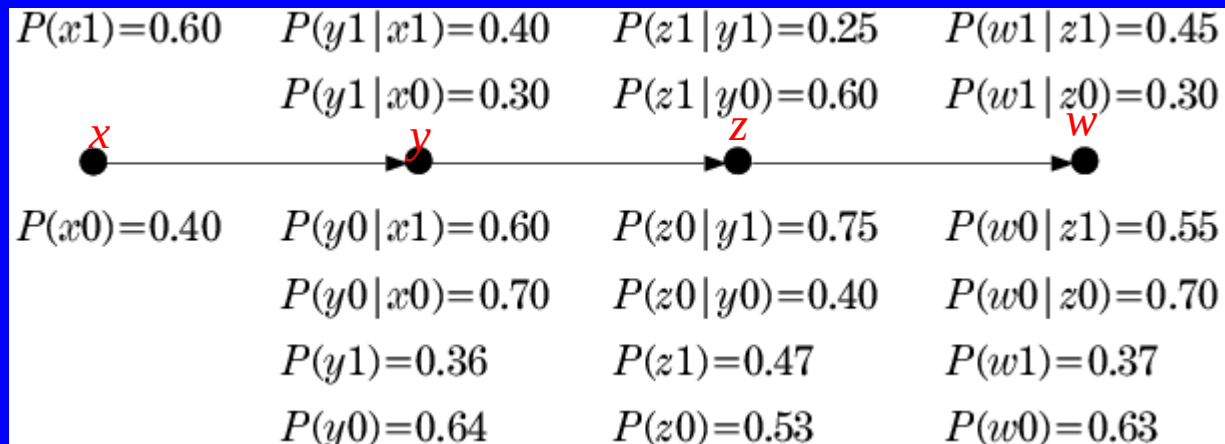
- Given this info, we can answer any probabilistic query:



a) If x is measured to be $x=1$ (x_1), compute $P(z_1|x_1)$ and $P(w_0|x_1)$.

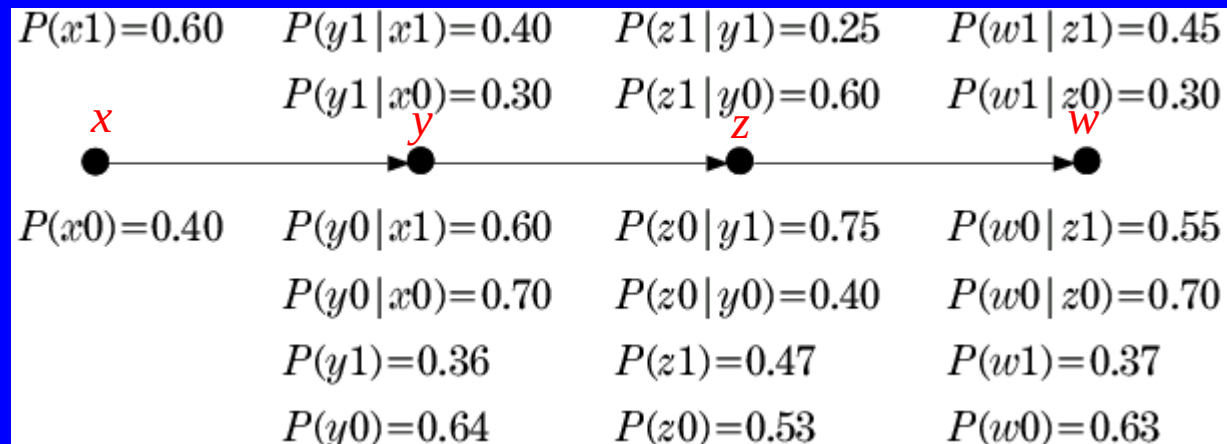
b) If w is measured to be $w=1$ (w_1) compute $P(z_1|w_1)$.

a) If x is measured to be $x=1$ (x_1), compute $P(z_1|x_1)$ and $P(w_0|x_1)$.



$$\begin{aligned}
 P(z_1|x_1) &= P(z_1|y_1, x_1)P(y_1|x_1) + P(z_1|y_0, x_1)P(y_0|x_1) \\
 &= P(z_1|y_1)P(y_1|x_1) + P(z_1|y_0)P(y_0|x_1) \\
 &= (0.25)(0.4) + (0.6)(0.6) = 0.46
 \end{aligned}$$

a) If x is measured to be $x=1$ (x_1), compute $P(z_1|x_1)$ and $P(w_0|x_1)$.



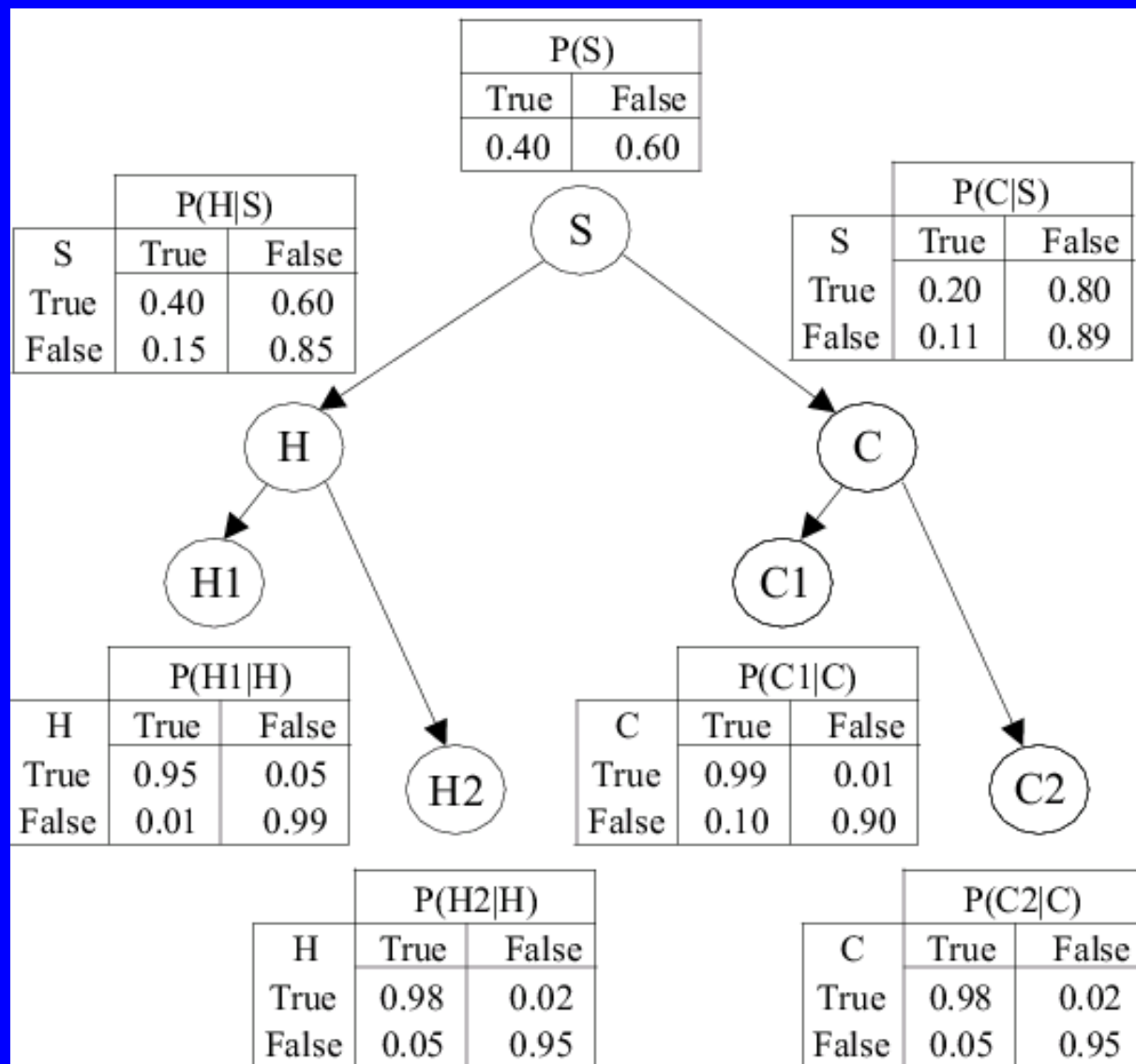
$$\begin{aligned}
 P(w_0|x_1) &= P(w_0|z_1, x_1)P(z_1|x_1) + P(w_0|z_0, x_1)P(z_0|x_1) \\
 &= P(w_0|z_1)P(z_1|x_1) + P(w_0|z_0)P(z_0|x_1) \\
 &= (0.55)(0.46) + (0.7)(0.54) = 0.63
 \end{aligned}$$

b) If w is measured to be $w=1$ (w_1) compute $P(z_1|w_1)$.

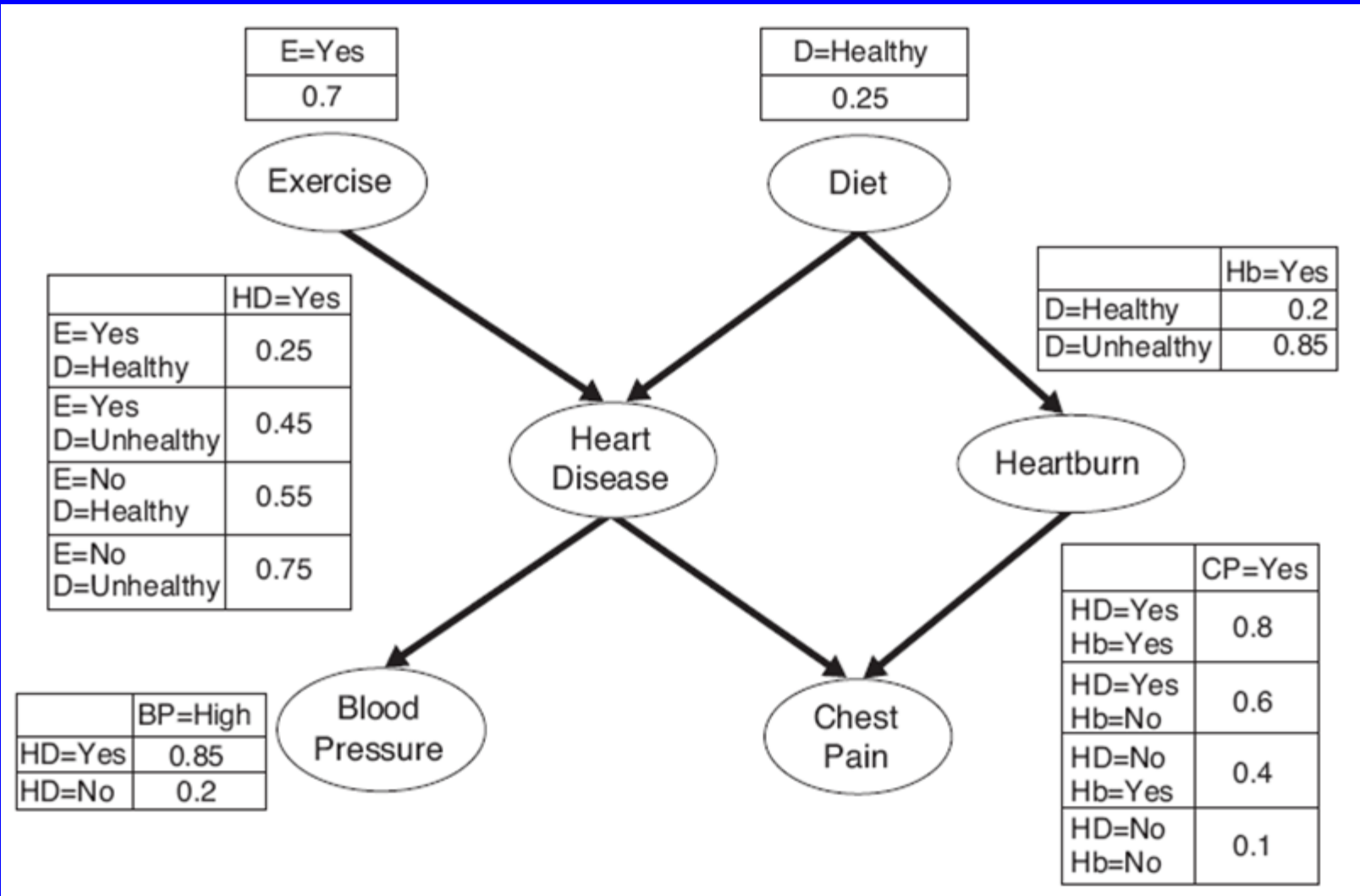
$P(x_1)=0.60$	$P(y_1 x_1)=0.40$	$P(z_1 y_1)=0.25$	$P(w_1 z_1)=0.45$
	$P(y_1 x_0)=0.30$	$P(z_1 y_0)=0.60$	$P(w_1 z_0)=0.30$
			
$P(x_0)=0.40$	$P(y_0 x_1)=0.60$	$P(z_0 y_1)=0.75$	$P(w_0 z_1)=0.55$
	$P(y_0 x_0)=0.70$	$P(z_0 y_0)=0.40$	$P(w_0 z_0)=0.70$
	$P(y_1)=0.36$	$P(z_1)=0.47$	$P(w_1)=0.37$
	$P(y_0)=0.64$	$P(z_0)=0.53$	$P(w_0)=0.63$

$$P(z_1|w_1) = \frac{P(w_1|z_1)P(z_1)}{P(w_1)} = \frac{(0.45)(0.47)}{0.37} = 0.57$$

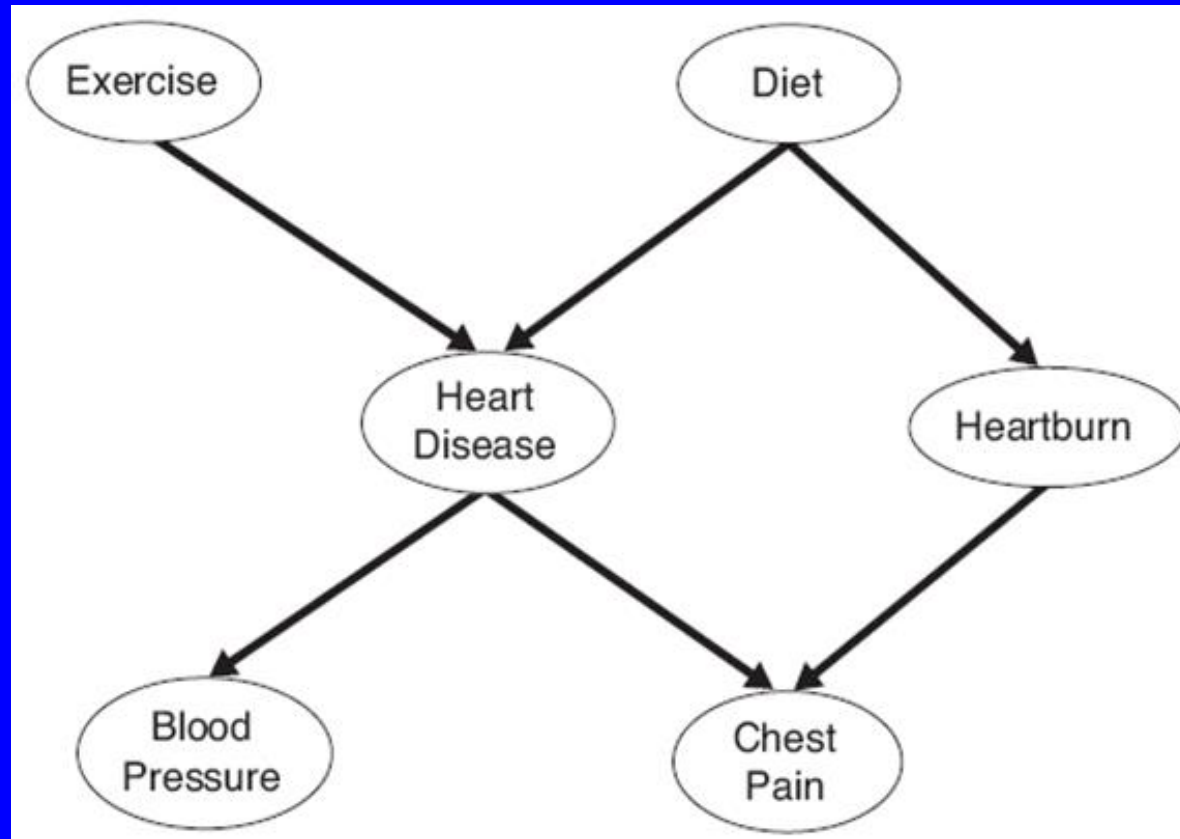
What's about more complex networks?



What's about more complex networks?

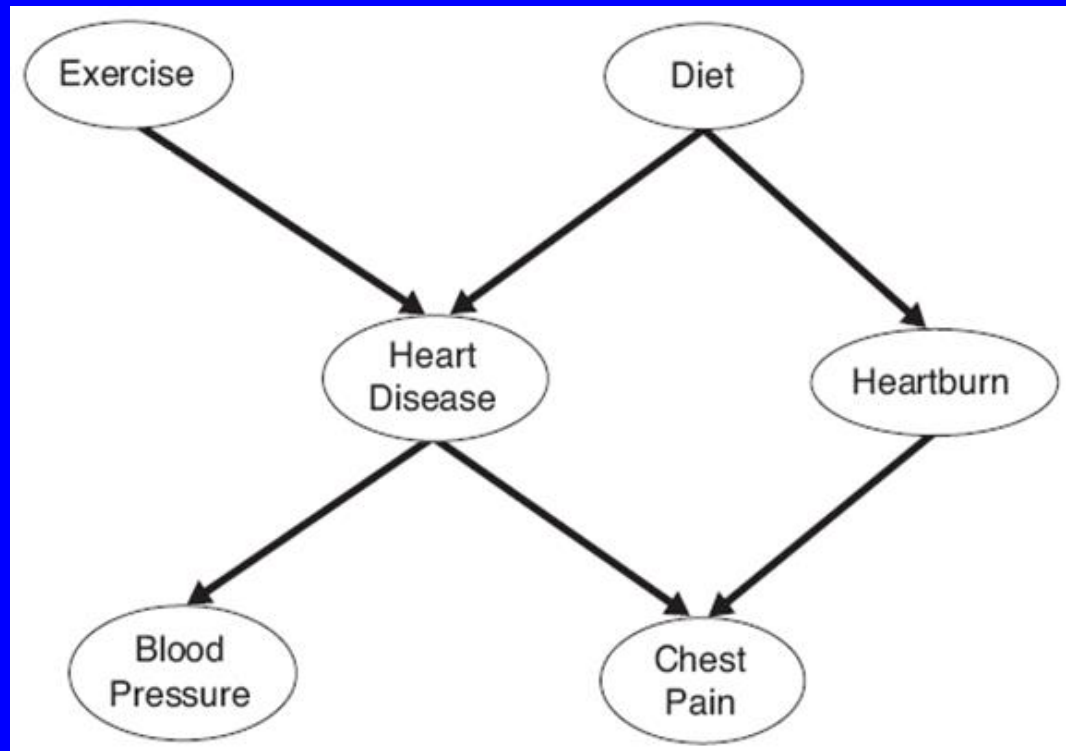


We will study this graph



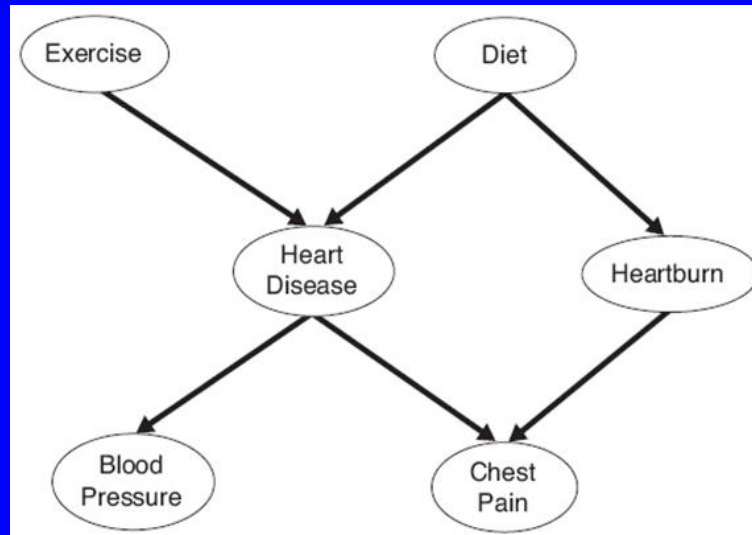
We can show:

- $P(D|E)=P(D)$
- $P(Hb|HD, E, D)=P(Hb|D)$
- $P(CP|Hb, HD, E, D)=P(CP|Hb, HD)$
- $P(BP|CP, Hb, HD, E, D)=P(BP|HD)$
- However, $P(HD|E,D)$ cannot be simplified



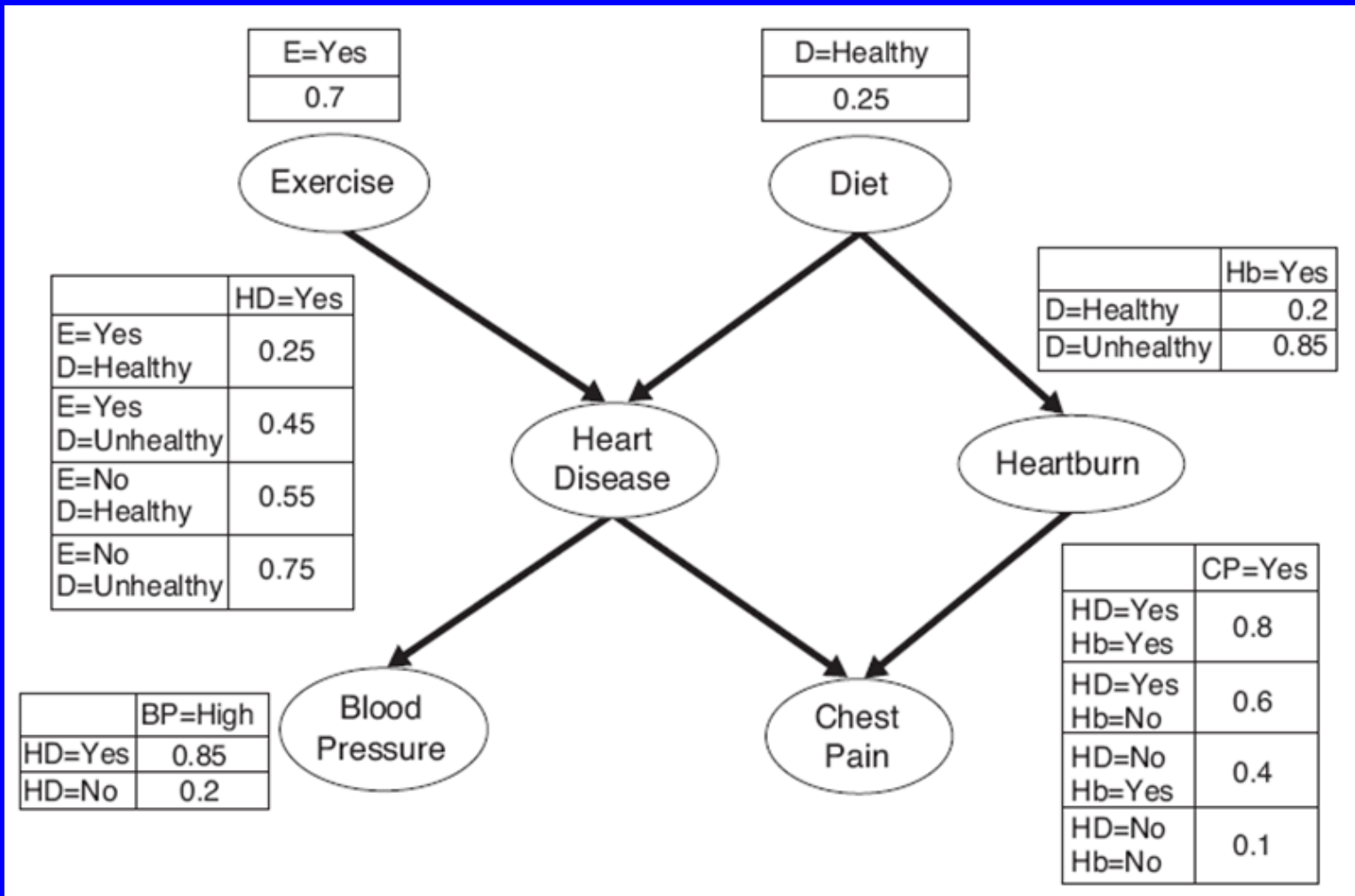
Exercise:

- $P(CP|HD, BP, E, D) = ?$



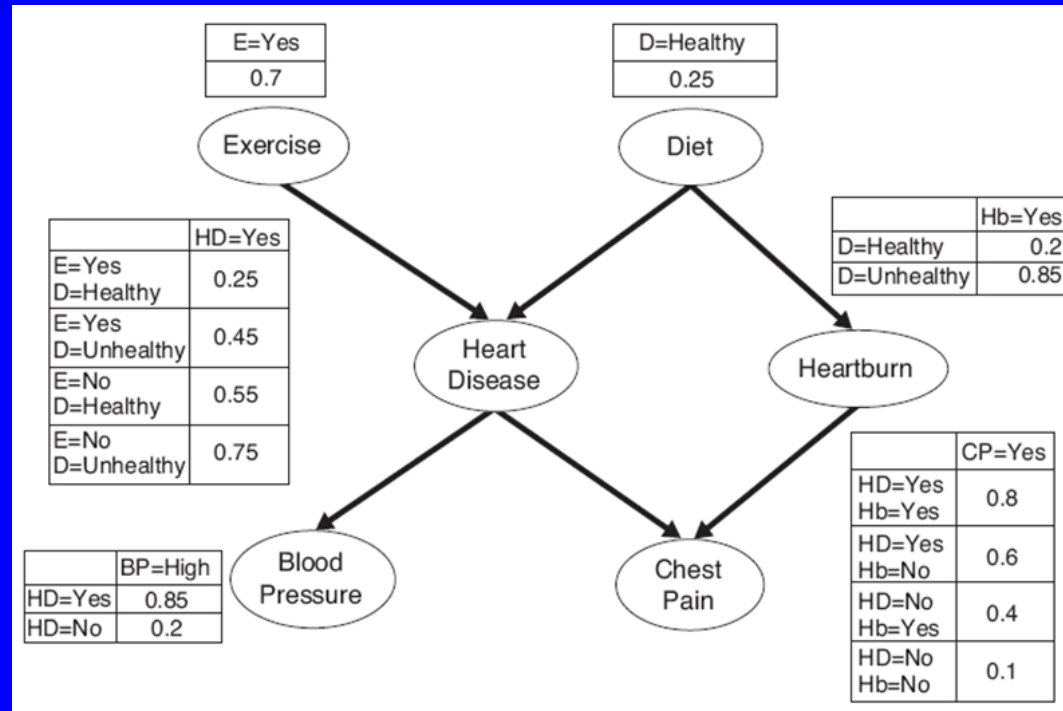
Exercise:

- $P(CP|HD, BP, E, D)$ = No simplification



Calculate $P(\text{HD=yes})$?

Calculate $P(HD=yes)$?



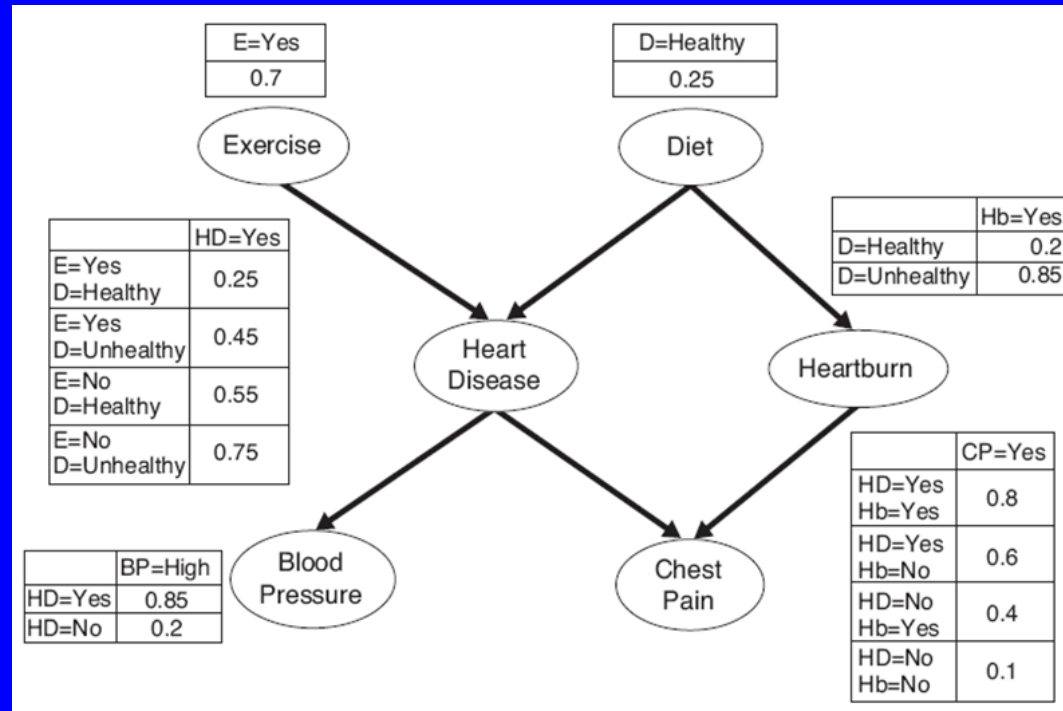
$$P(HD = Yes) = \sum_{\alpha} \sum_{\beta} P(HD = yes | E = \alpha, D = \beta) P(E = \alpha, D = \beta)$$

where,

α = Set of Values of Exercise(E) = {Yes, No}

β = Set of Values of Diet(D) = {Healthy, Not Healthy}

Calculate $P(HD=yes)$?

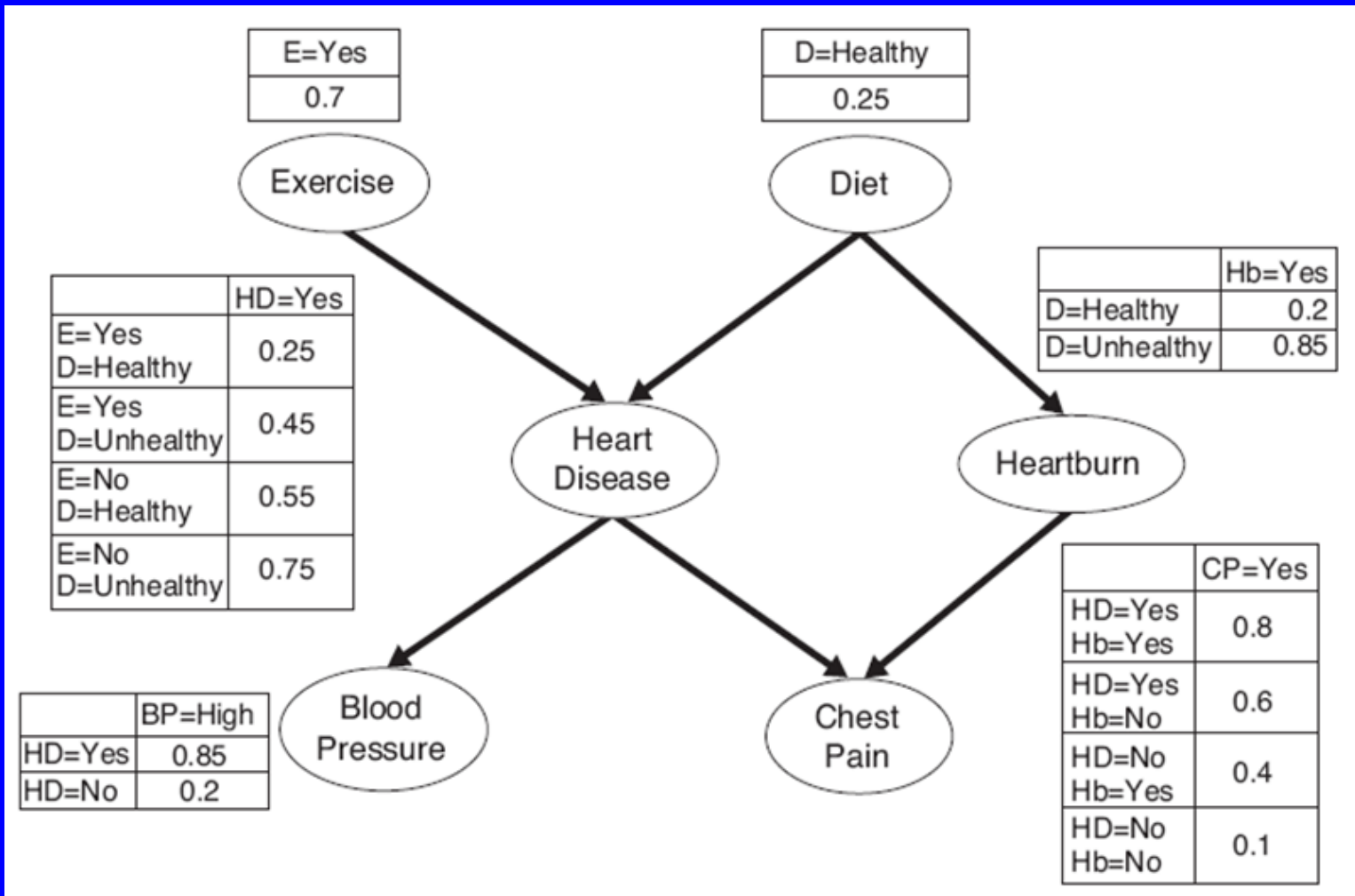


$$P(HD = Yes) = \sum_{\alpha} \sum_{\beta} P(HD = yes | E = \alpha, D = \beta) P(E = \alpha, D = \beta)$$

$$= \sum_{\alpha} \sum_{\beta} P(HD = yes | E = \alpha, D = \beta) P(E = \alpha) P(D = \beta)$$

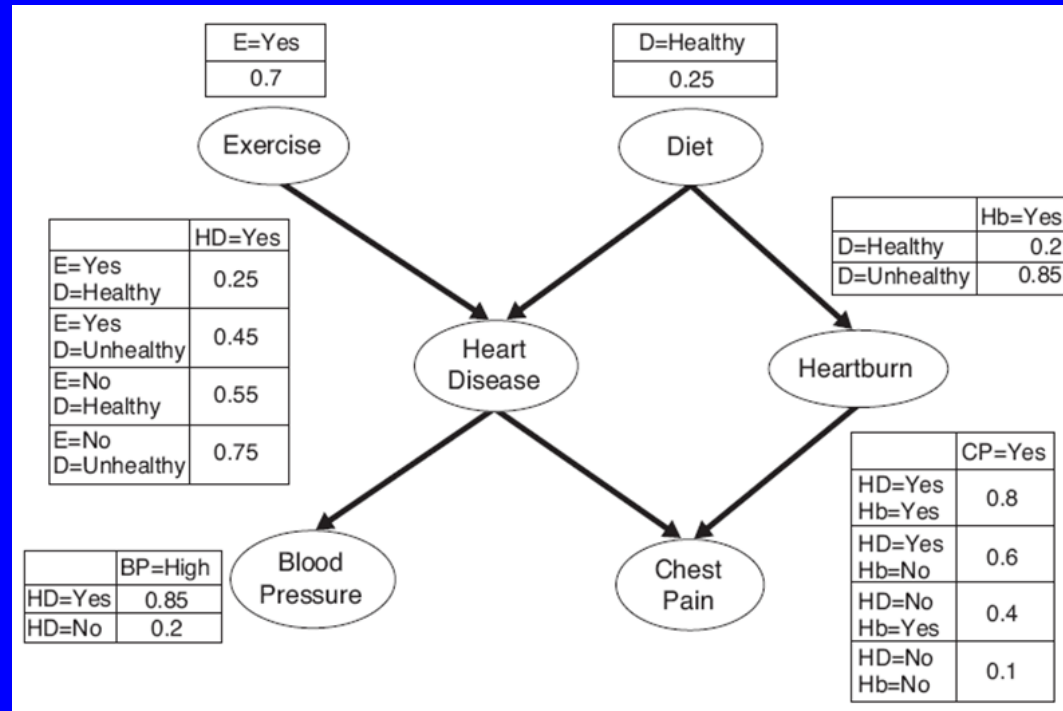
$$= 0.25 \times 0.7 \times 0.25 + 0.45 \times 0.7 \times 0.75 + 0.55 \times 0.3 \times 0.25 + 0.75 \times 0.3 \times 0.75$$

$$= 0.49$$



Calculate $P(\text{HD=yes} | \text{BP=High})$?

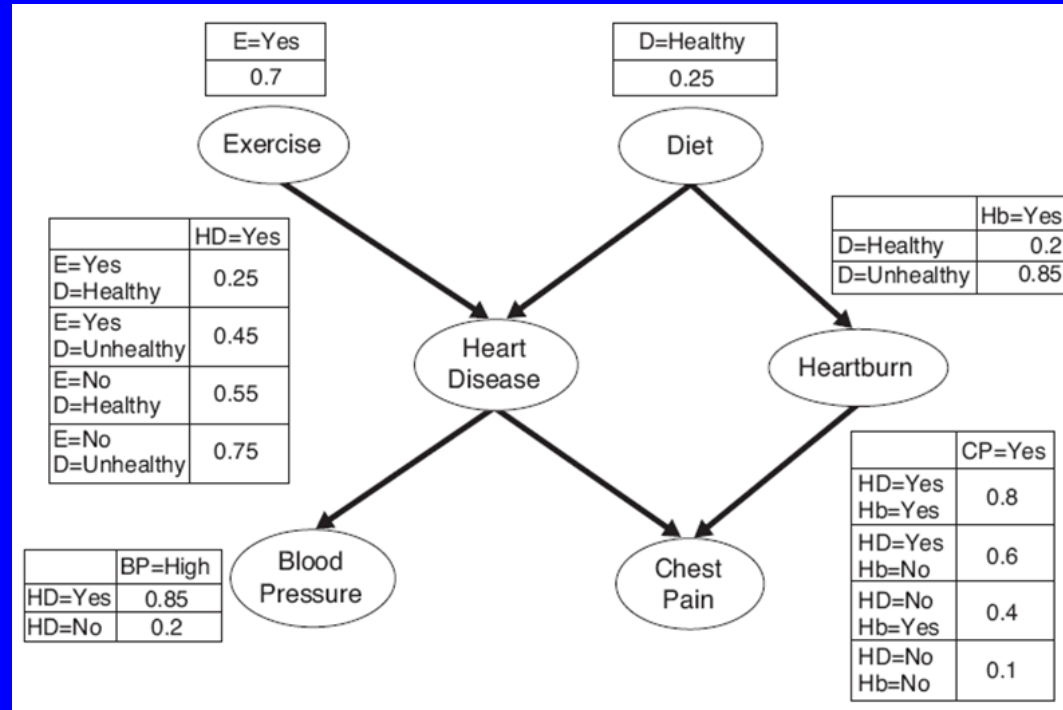
Calculate $P(HD=yes | BP=High)$



$P(HD = yes | BP = High)$ can be written as

$$\frac{P(BP = High | HD = yes)P(HD = yes)}{P(BP = High)}$$

Calculate $P(HD=yes | BP=High)$

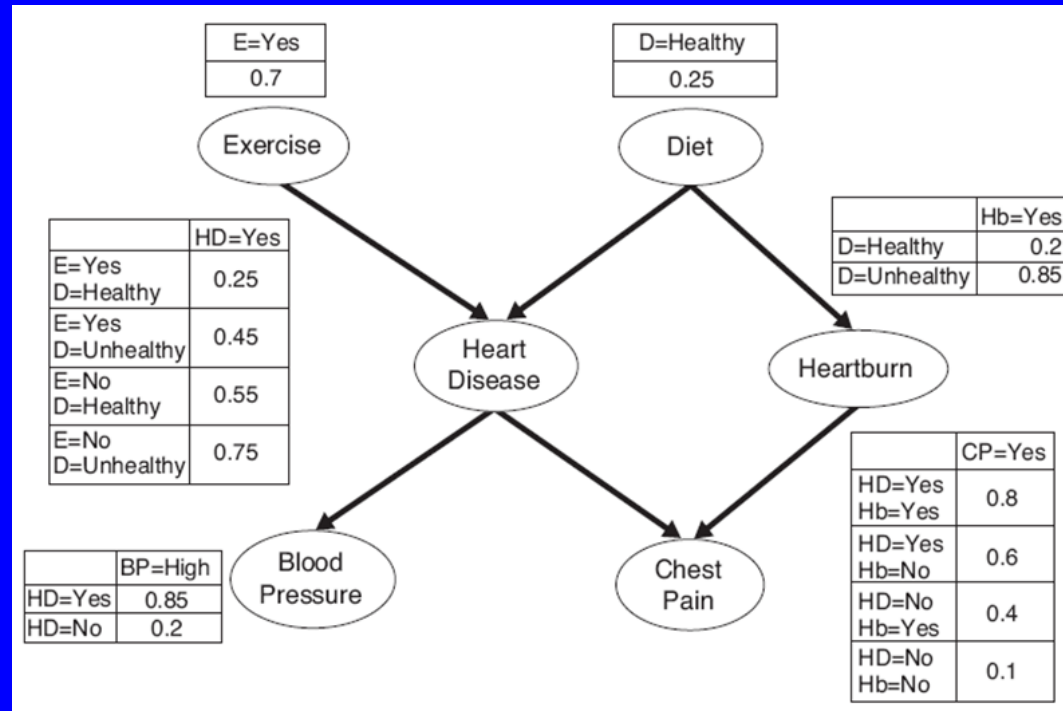


$$P(BP = High) = \sum_{\gamma} P(BP = high | HD = \gamma) P(HD = \gamma)$$

where,

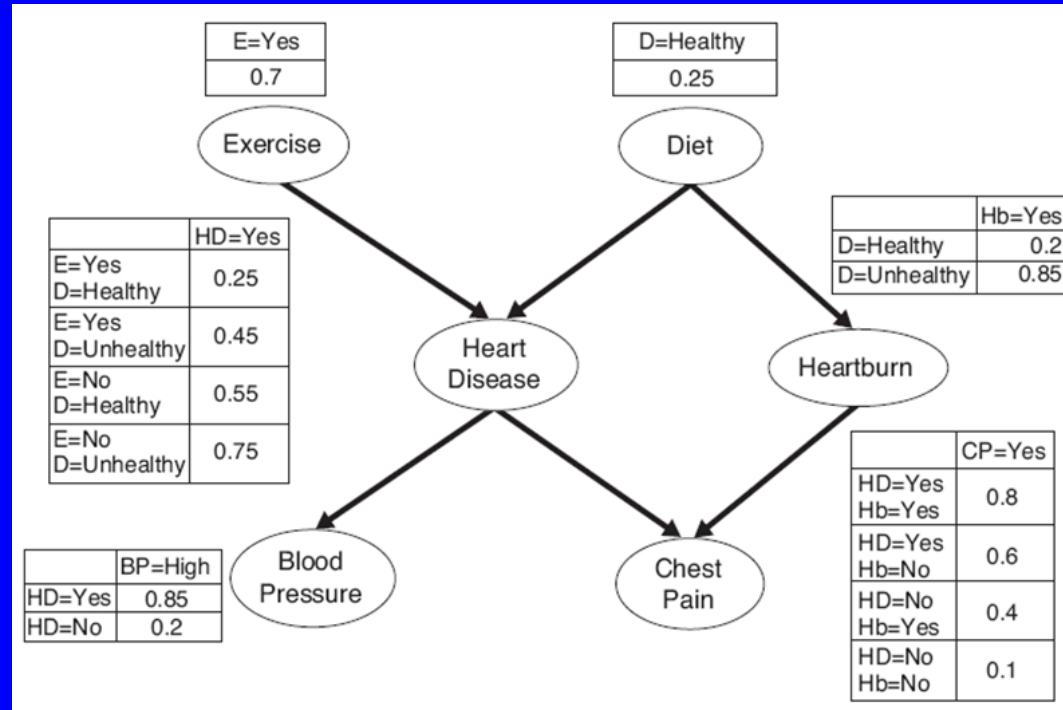
γ = Set of Values of Heart Disease (HD) = {Yes, No}

Calculate $P(HD=yes | BP=High)$

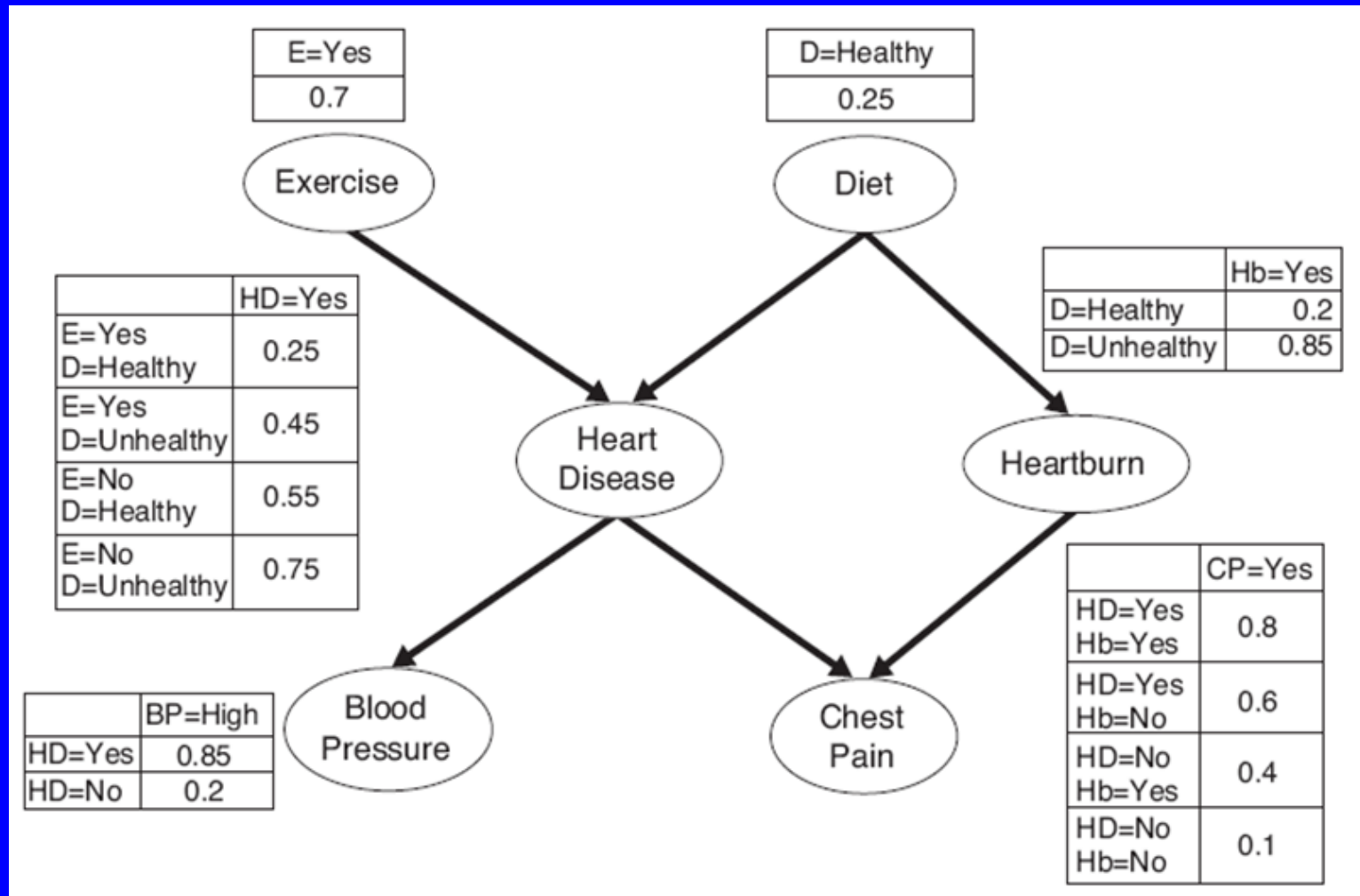


$$\begin{aligned}
 P(BP = High) &= \sum_{\gamma} P(BP = high | HD = \gamma) P(HD = \gamma) \\
 &= 0.85 \times 0.49 + 0.2 \times 0.51 = 0.5185
 \end{aligned}$$

Calculate $P(HD=yes | BP=High)$

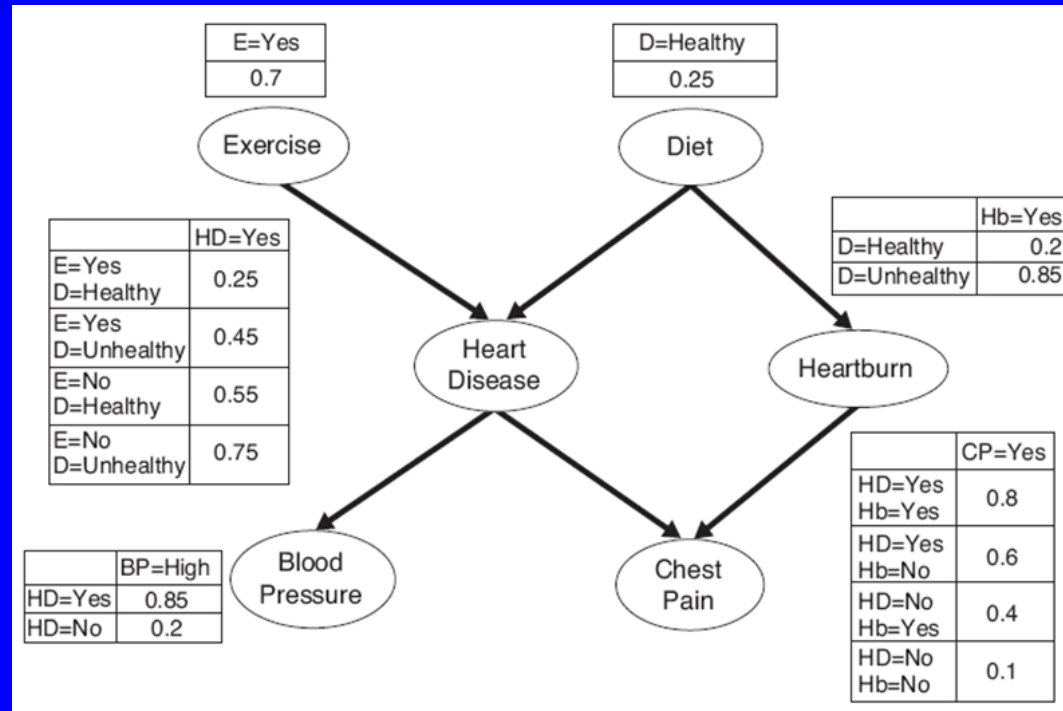


$$\begin{aligned}
 P(HD = yes | BP = High) &= \frac{P(BP = High | HD = yes)P(HD = yes)}{P(BP = High)} \\
 &= \frac{0.85 \times 0.49}{0.5185} = 0.8033
 \end{aligned}$$



Calculate $P(\text{HD=yes} | \text{BP=high}, \text{D=Healthy}, \text{E=yes})$?

Calculate $P(HD=yes | BP=high, D=Healthy, E=yes)$?



$$\begin{aligned}
 &P(HD = yes | BP = high, D = Healthy, E = Yes) \\
 &= \frac{P(BP = high | HD = yes, D = Healthy, E = Yes)}{P(BP = high | D = Healthy, E = Yes)} \times P(HD = yes | D = Healthy, E = Yes)
 \end{aligned}$$

How is this formula true?

$$P(HD = yes | BP = high, D = Healthy, E = Yes) \\ = \frac{P(BP = high | HD = yes, D = Healthy, E = Yes)}{P(BP = high | D = Healthy, E = Yes)} \times P(HD = yes | D = Healthy, E = Yes)$$

Let

$$P(X | Y) = \frac{P(Y | X)}{P(Y)} \times P(X)$$

Now add Z and W as condition

$$P(X | Y, Z, W) = \frac{P(Y | X, Z, W)}{P(Y | Z, W)} \times P(X | Z, W)$$

Similarly,

$$P(HD = yes | BP = high) = \frac{P(BP = high | HD = yes)}{P(BP = high)} \times P(HD = yes)$$

Now add conditions $D = Healthy$ and $E = Yes$ to above formula

Similarly,

$$\begin{aligned} &P(BP = high \mid D = Healthy, E = Yes) \\ &= \sum_{\gamma} P(BP = high \mid HD = \gamma, D = Healthy, E = Yes) \times P(HD = \gamma \mid D = Healthy, E = Yes) \end{aligned}$$

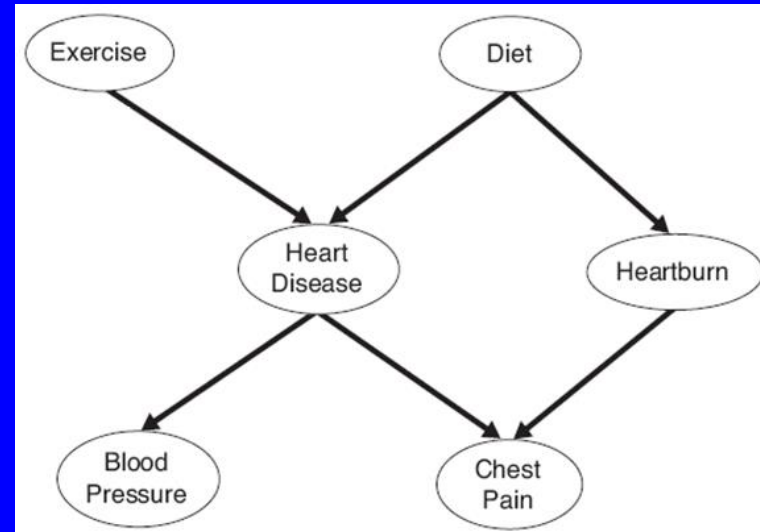
Proof:

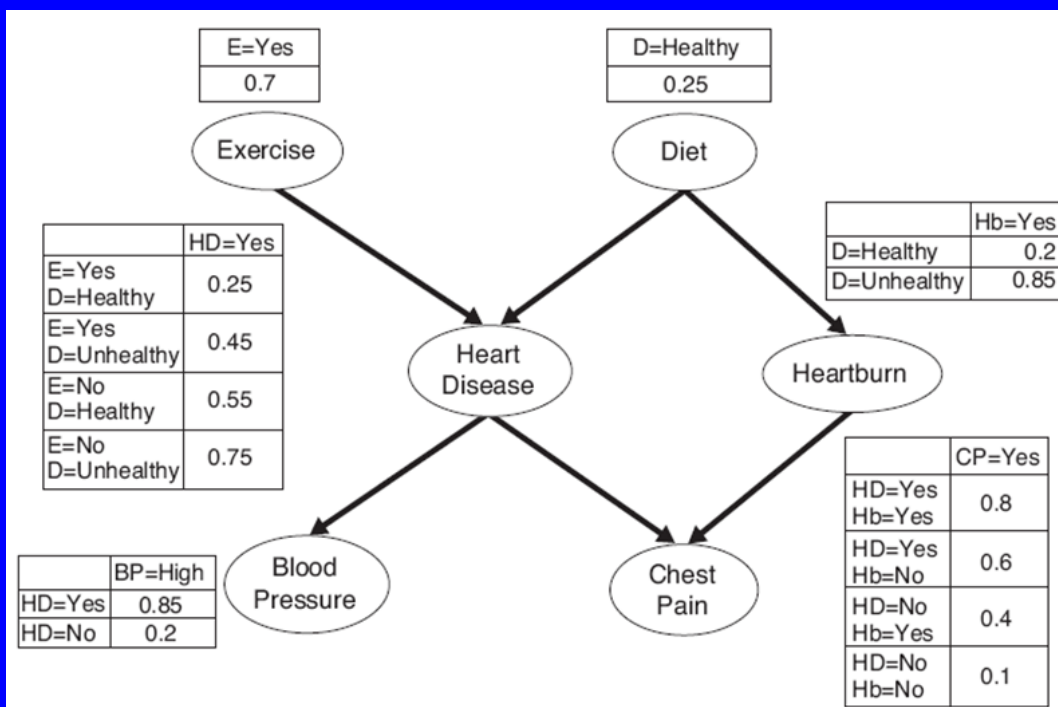
$$P(BP = high) = \sum_{\gamma} P(BP = high \mid HD = \gamma) \times P(HD = \gamma)$$

Adding conditions $D = Healthy$ and $E = Yes$

we get,

$$\begin{aligned} &P(BP = high \mid D = Healthy, E = Yes) \\ &= \sum_{\gamma} P(BP = high \mid HD = \gamma, D = Healthy, E = Yes) \times P(HD = \gamma \mid D = Healthy, E = Yes) \\ &= \sum_{\gamma} P(BP = high \mid HD = \gamma) \times P(HD = \gamma \mid D = Healthy, E = Yes) \end{aligned}$$





$$\begin{aligned}
 &P(HD = \text{yes} \mid BP = \text{high}, D = \text{Healthy}, E = \text{Yes}) \\
 &= \frac{P(BP = \text{high} \mid HD = \text{yes}, D = \text{Healthy}, E = \text{Yes})}{P(BP = \text{high} \mid D = \text{Healthy}, E = \text{Yes})} \times P(HD = \text{yes} \mid D = \text{Healthy}, E = \text{Yes}) \\
 &= \frac{P(BP = \text{high} \mid HD = \text{yes})}{P(BP = \text{high} \mid D = \text{Healthy}, E = \text{Yes})} \times P(HD = \text{yes} \mid D = \text{Healthy}, E = \text{Yes}) \\
 &= \frac{P(BP = \text{high} \mid HD = \text{yes})}{\sum_{\gamma} P(BP = \text{high} \mid HD = \gamma) P(HD = \gamma \mid D = \text{Healthy}, E = \text{Yes})} \times P(HD = \text{yes} \mid D = \text{Healthy}, E = \text{Yes}) \\
 &= \frac{0.85 \times 0.25}{0.85 \times 0.25 + 0.2 \times 0.75} = 0.5862
 \end{aligned}$$